Aadit Pani

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SUMMARY

AI/ML and security-focused Computer Science student with hands-on experience in **edge-based intrusion detection**, adversarial ML, and **agentic AI systems**. Built production-grade projects including a **few-shot IDS using Prototypical Networks**, **GAN-based attack synthesis pipelines**, **NeuroSym-AI** (a neuro-symbolic guardrail library on PyPI), and **NORA** (a local voice-controlled AI operating system with screen intelligence and cognitive memory). Strong builder combining ML modeling, systems engineering, and open-source delivery.

EDUCATION

RV University, Bengaluru — Sept 2023 – June 2027
B.Tech (Hons.) in Computer Science (AI/ML), Minor in FinTech
GPA: 8.2 / 10

EXPERIENCE

HYRGPT — AI Engineering Intern (Healthcare AI) — Feb 2026 – Present

- Contributed to a **production-grade healthcare AI assistant** for PCOS awareness and symptom guidance, building Python backend pipelines integrating LLM reasoning with **external knowledge retrieval**.
- Designed a **PCOS knowledge pipeline** combining curated medical resources, retrieval systems, and prompt orchestration to improve reliability of health recommendations.
- Implemented **safety layers** including response validation, structured outputs, and guardrails to prevent unsafe medical advice; developed API services supporting **real-time conversational healthcare interactions**.

PUBLICATIONS

- **C4PS: An Interactive Pipeline for Social Media Image Enhancement and Multilingual Captioning** — Accepted, IEEE CAI 2026

SKILLS

AI & Security: Intrusion Detection Systems (IDS), GANs, CNNs, Few-Shot Learning, Prototypical Networks, Threat Modeling, Explainability, Quantization, TorchScript
Neuro-Symbolic & Guardrails: Symbolic Policy Rules, JSON Schema Validation, Regex Policies, Composite Policy Algebra, Repair Loops, Audit Traces, Z3 (optional)
Programming: Python, C, C++, JavaScript
Backend & Systems: Flask, SQLite3, Docker, Kubernetes, GitHub Actions
Edge & IoT: Raspberry Pi, ESP32, Arduino, MQTT, GPS, PIR/Ultrasonic Sensors, Secure Edge Deployment
FinTech & Data: Fraud Analytics, Digital Payments, Blockchain Fundamentals, GCP BigQuery
Tools: Git, VS Code, NumPy, Matplotlib, ChromaDB, faster-whisper, Google Colab

KEY PROJECTS

NORA — Local Voice AI Operating System (Personal Project) — 2026 – Present

- Built a **fully local, voice-controlled AI assistant** on Windows: faster-whisper STT → NeuroSym input guard → LLM intent parser (Groq / Claude / Ollama) → command engine → edge-TTS, processing voice commands end-to-end with no cloud dependency.
- Implemented a **Screen Intelligence engine** using Claude Vision (Haiku) for natural-language UI interaction: describe screen contents, click UI elements by voice description, OCR extraction, background condition monitoring (*"watch for the training to finish"*), and automated error/stack-trace diagnosis.
- Designed a **Cognitive Memory System v2** using ChromaDB and sentence-transformers for semantic recall and episodic memory, paired with a **Proactive Intelligence Engine** that predicts user workflows from time-of-day patterns and bigram command sequences.
- Integrated **NeuroSym-AI** as the primary security layer: prompt injection guards on raw transcriptions (9 attack categories) and action-level policy enforcement (destructive confirmation, path sandbox, step-count limits).
- Supports **40+ registered voice commands** spanning OS automation, developer tooling (git, pytest, pip), music, focus/productivity, and semantic memory.

NeuroSym-AI — Neuro-Symbolic Guardrails for LLMs (Open-Source, PyPI) — Jul 2025 – Present

- Designed a **provider-agnostic neuro-symbolic guardrail engine** covering the full pipeline — voice inputs, LLM outputs, and agent action plans — without requiring API keys or cloud calls.
- Built a **PromptInjectionRule spanning 9 attack categories** (path traversal, role switch, obfuscation, delimiter injection, exfiltration, and more); validated on a **built-in 134-case adversarial benchmark** achieving **79.8% block rate, 0% false positives, and 0.48 ms average latency**.
- Implemented a **boolean policy algebra** (AllOf, AnyOf, Not, Implies) for composing symbolic validation layers: Regex policies, JSON Schema enforcement, Python predicate rules, and action-graph policies, each with full **structured audit traces**.
- Integrated an optional **LLM repair loop** with fallback routing (Ollama, Gemini) while keeping the core execution path deterministic and fully offline.
- Packaged and released as an **open-source PyPI library** (v0.2.0) with strict typing (**py.typed**), Typer CLI, and mypy strict mode. Available at `pip install neurosym-ai`

EdgeGenSec — Lightweight Intrusion Detection System (AI/ML Internship) Jul 2025 – Present

- Built a **real-time multiclass IDS** for Raspberry Pi 5 using **78D CICIDS-style flow features**.
- Generated rare attack samples (*SQLi, XSS, Heartbleed, Infiltration*) using **GANs**, improving recall on underrepresented classes.
- Implemented **few-shot learning with Prototypical Networks**, outperforming CNN baselines for low-data attack detection.
- Optimized inference using **TorchScript and quantization**, reducing latency and memory footprint for edge deployment.
- Added an **adaptive retraining loop** to support continuous learning on uncertain traffic patterns.

Geo-Fence Swarm — Autonomous Perimeter Monitoring (IoT + Security + ML) Sep 2025 – Present

- Architected a **distributed Raspberry Pi + ESP32 security swarm** for autonomous perimeter monitoring, integrating PIR motion sensors, ultrasonic sensors, and GPS modules via GPIO for multi-layer intrusion detection.
- Built a **motor-controlled rover platform** with L298N drivers and real-time control logic for autonomous patrol, obstacle avoidance, and geo-boundary navigation.
- Deployed **lightweight YOLO inference on-device** for human detection with automatic image capture, classification, and geo-tagged event logging.
- Designed a **secure MQTT communication layer** for real-time telemetry between rover and static nodes, with a Flask-based streaming dashboard for remote monitoring.

C4PS — Secure Captioning, Enhancement & Multilingual Processing Jul 2025 – Present

- Built a secure multimedia pipeline using Real-ESRGAN, transformer-based captioning, and multilingual translation; accepted at **IEEE CAI 2026**.
- Implemented authenticated APIs, token-scoped access, and auditable media-processing workflows.

Hybrid CNN Initialization Research — Orthogonal + He Initialization 2024 – Present

- Proposed a **hybrid initialization method** combining Orthogonal and He initialization across ResNet, VGG, and GoogLeNet, achieving faster convergence and improved gradient stability over standard baselines.

OTHER PROJECTS

- Habitat Assessment Rover — modular environmental rover with telemetry and fault-tolerant sensing.
- Movie Recommender System — Flask + SQLite3 with collaborative filtering.

ACHIEVEMENTS & INTERESTS

- Co-authored research on **hybrid CNN initialization** and ML security techniques; IEEE CAI 2026 publication accepted.
- Interests: AI security, voice AI systems, edge robotics, fraud analytics, secure MLOps.